

2 details to recap: ① Bias variance tradeoff ② Classification

2 goals:

ch3: Linear regression basics

Recap:

$$Y = \underbrace{f(X)}_{\text{data}} + \underbrace{\epsilon}_{\text{error}} \quad \left(\begin{array}{l} \text{statistical} \\ \text{learning} \end{array} \right. \left. \begin{array}{l} \text{framework} \\ \text{regression case} \end{array} \right)$$

$$f(X) \approx \hat{f}(X) \quad (Y \approx \hat{f}(X) = \hat{Y})$$

fit to train data

Aim to minimize

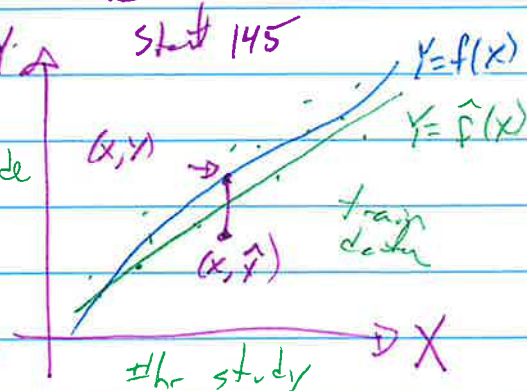
$(f, \hat{f})$
fixed

$$E[(Y - \hat{Y})^2] = \underbrace{(f(X) - \hat{f}(X))^2}_{\text{Red error}} + \text{Var}(\epsilon)$$

avg squared error

Red error

Irred error

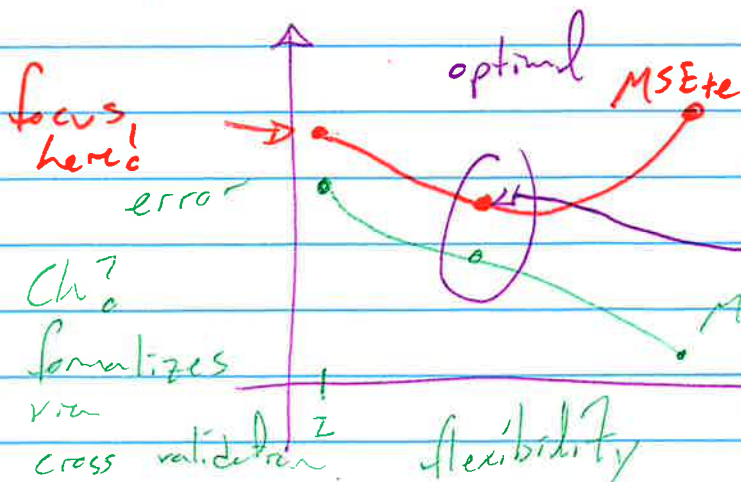


$f \approx \hat{f}$, measure goal?

① Train data to fit $\hat{f}$ to (X_i, Y_i) $i=1, \dots, n$

② Test data, unseen by $\hat{f}$ (X_{oi}, Y_{oi}) $i=1, \dots, k$

Over fit & unfit tradeoff



$$MSE_{tr} = \frac{1}{n} \sum_i (Y_i - \hat{Y}_i)^2$$

$$MSE_{te} = \frac{1}{k} \sum_i (Y_{oi} - \hat{Y}_{oi})^2$$

Minimize MSE_{te}

$$\mu = E[X] = \sum p(x) x$$

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$E[X_1 + X_2] = E[X_1] + E[X_2] \quad \text{pg 2}$$

$$E[cX] = cE[X]$$

Bias - Variance tradeoff (test data accuracy)

$$MSE_{\text{test}} = E[(Y_0 - \hat{f}(x_0))^2] = \underbrace{\text{Var}(\hat{f}(x_0))}_{\substack{\hat{f} \text{ variance} \\ \text{on diff train} \\ \text{sets}}} + \underbrace{[\text{Bias}(\hat{f}(x_0))]^2}_{\substack{\text{"Bias" of} \\ \hat{f}}} + \underbrace{\text{Var}(\epsilon)}_{\text{irr error}}$$

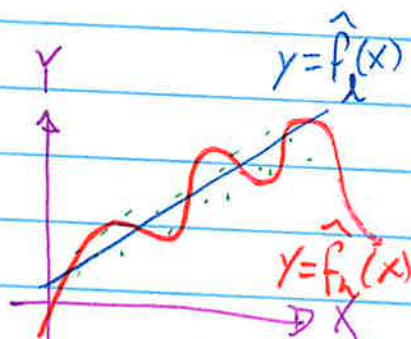
mean test data error

Assume test data fixed & f fixed

$$\text{Bias}(\hat{f}(x_0)) = \underbrace{E[\hat{f}(x_0)]}_{\text{Avg pred}} - f(x_0)$$

$\text{Var}(\hat{f}(x_0))$ measures how much $\hat{f}$ changes with new training data

	$\text{Var}(\hat{f}(x_0))$	$\text{Bias}(\hat{f}(x_0))$
Low flex	Low	High
High flex	High	Low



Line always has same shape

High flex adapts with train data

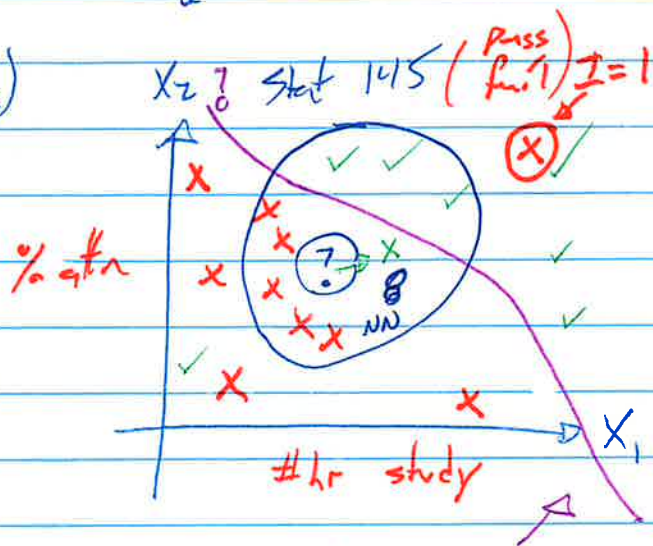
Minimizing MSE_{test} wants both variance & bias low, but not usually possible.

Classification Intro: Text full details, Ch4 revisits.

Model discrete groups (classes)

$$Y = f(X) + \epsilon$$

data $\rightarrow$ f unknown f $\rightarrow$ error $\rightarrow$ pass/fail



How to assess $f \approx \hat{f}$?

$\rightarrow$ Percent accuracy.

Indicator

learn a boundary

(Test set similar)

Training set error rate

$$= \frac{1}{n} \sum \mathbb{I}(y_i \neq \hat{y}_i)$$

n train size $\rightarrow$ (perc decision misclassified) f

If classify wrong.

unknown formula

$$\mathbb{I}(y_i \neq \hat{y}_i) = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

2 basic classifiers: ① Bayes ($P(Y=y_i | X=x_i)$) $\rightarrow$ Maxes $\checkmark$
② K-nearest neighbors (KNN)

Bayes classifier is best performing model, but still has irreducible error, but not usable in practice b/c $P(Y=y_i | X=x_i)$ is not computable.

KNN $P(Y=y_i | X=x_i) \approx \frac{1}{K} \sum_{i \in N_0} \mathbb{I}(y_i = i)$

Perc of neighbors in class i

k nearest neighbors

Choose best K (K small, highly flexible)
 (K big, low flexible)

Train-test & bias-variance tradeoffs
 both occur in same way as regression.

Chapter 3 Linear regression

Simple linear regression.

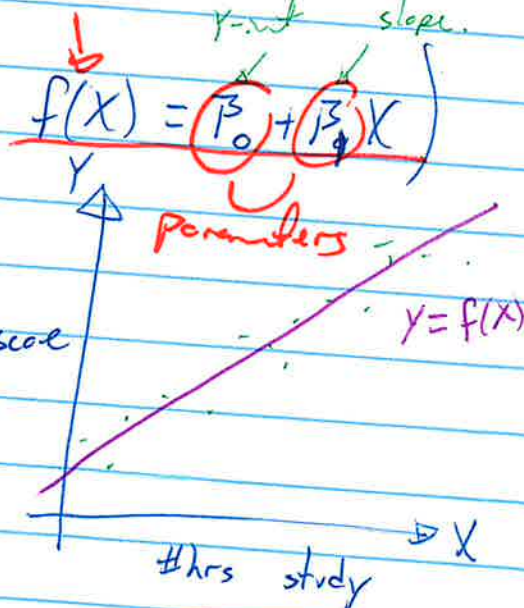
W2 (Simple LR, 2D)
 W3 (Multiple LR, 3+D)
 Parameter & extensions

$Y = f(X) + \epsilon$ (assume

$\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ so

$\hat{\beta}_0 \approx \beta_0$ & $\hat{\beta}_1 \approx \beta_1$

To get $\hat{f}$, choose $\hat{\beta}_0$ & $\hat{\beta}_1$
 in $\hat{f}(x)$ which minimizes
 error.



Best fit to train data

$$e_i = y_i - \hat{y}_i \text{ (residual)}$$

Choose β_0, β_1 by minimizing

(residual sum of squares) $RSS = e_1^2 + e_2^2 + \dots + e_n^2$

$$= \sum_i e_i^2$$

$$= \sum_i (y_i - \hat{y}_i)^2 = \sum_i (y_i - (\beta_0 + \beta_1 x_i))^2$$

So $RSS = RSS(\beta_0, \beta_1)$

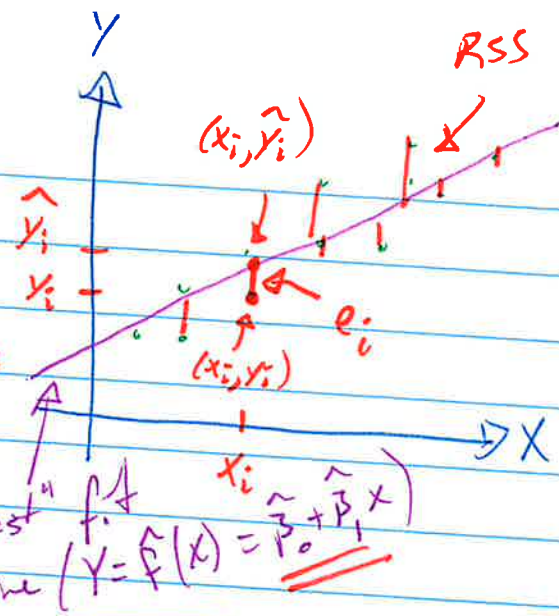
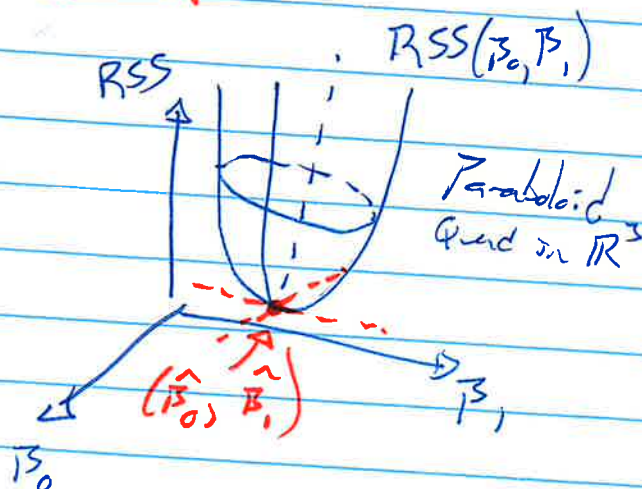
Want $(\hat{\beta}_0, \hat{\beta}_1)$ which minimize $RSS(\beta_0, \beta_1)$. i.e.

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{(\beta_0, \beta_1)}{\operatorname{argmin}} RSS(\beta_0, \beta_1)$$

For simple linear regression, use Calc to get formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$.

$$\frac{\partial RSS}{\partial \beta_0} = 0 \rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\frac{\partial RSS}{\partial \beta_1} = 0 \rightarrow \hat{\beta}_1 = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)}$$



$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X \quad (\text{Best fit line to train data})$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

Also can view LA as least squares. ($y = \beta_0 + \beta_1 x$)

$$\beta_0 + \beta_1 x_1 = y_1$$

$$\beta_0 + \beta_1 x_2 = y_2$$

$\vdots$

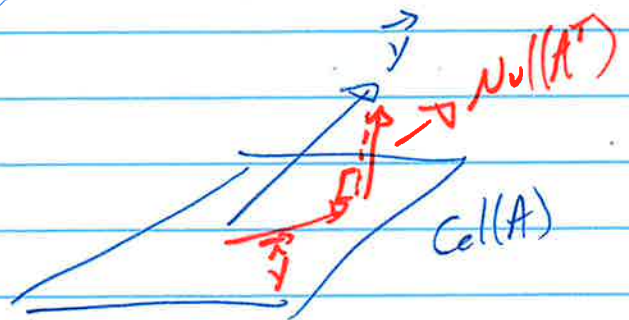
$$\beta_0 + \beta_1 x_n = y_n$$

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$A\vec{\beta} = \vec{y}$$

$$(A^T A)\vec{\beta} = A^T \vec{y}$$

$$\vec{\beta} = (A^T A)^{-1} A^T \vec{y}$$



one indep
var - x

Ch 3 Simple linear regression

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pg 1

$$Y = f(X) + \varepsilon = (\beta_0 + \beta_1 X) + \varepsilon$$

population
reg line.



Use training data (sample) to
fit reg. line.

$$f(X) = \beta_0 + \beta_1 X \approx \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X = \hat{Y}$$

Find best fit $\hat{\beta}_0 + \hat{\beta}_1$ by minimizing

$$RSS = \sum_i e_i^2 = \sum_i (y_i - \hat{y}_i)^2$$

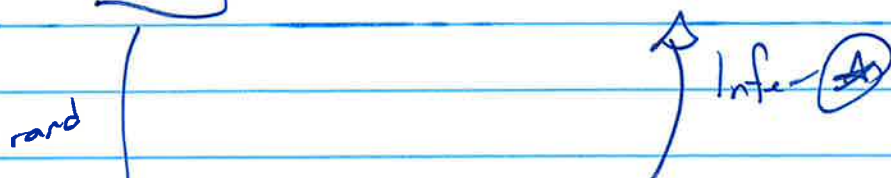
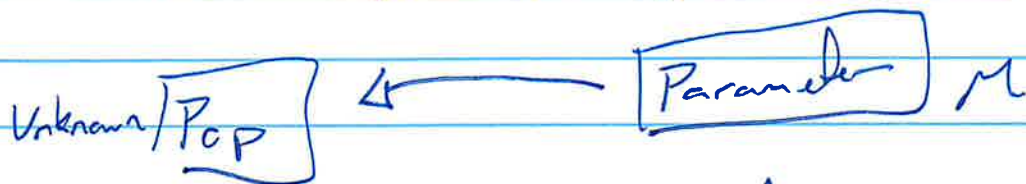
$$\checkmark \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Statistical inference of $\hat{\beta}_0, \hat{\beta}_1$ (certainty)
(CI/HT)

↳ Basic inference review. (Stat 145)

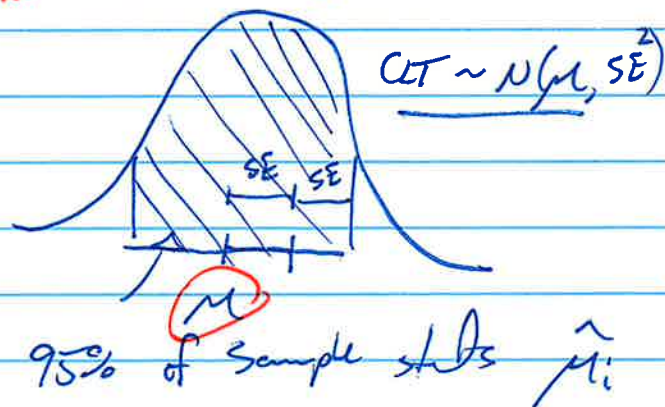
Stat inference for a avg of X | Exam avg



How sure am I that $\hat{\mu}$ is correct?

BE dist / Sampling distr

Many samples
 $\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_k$



Derive $SE(\mu)$ formula

$$Var(\mu) = SE(\mu)^2 = \left(\frac{\sigma}{\sqrt{n}}\right)^2 = \frac{\sigma^2}{n}$$

σ^2 standard dev.

only need 1 sample $\downarrow$

So 95% CI
 Sample stand dev.

$$\mu \pm 2SE$$

$$\left[\hat{\mu} - 2 \frac{s}{\sqrt{n}}, \hat{\mu} + 2 \frac{s}{\sqrt{n}} \right] \sim \left[\mu - 2SE, \mu + 2SE \right]$$

$$\left[\hat{\mu} - 2 \frac{\sigma}{\sqrt{n}}, \hat{\mu} + 2 \frac{\sigma}{\sqrt{n}} \right]$$

think $\hat{\mu} = 86$

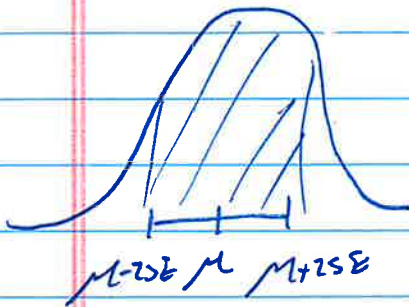
Hypothesis tests: $\hat{\mu} \approx \mu$

Shift sample

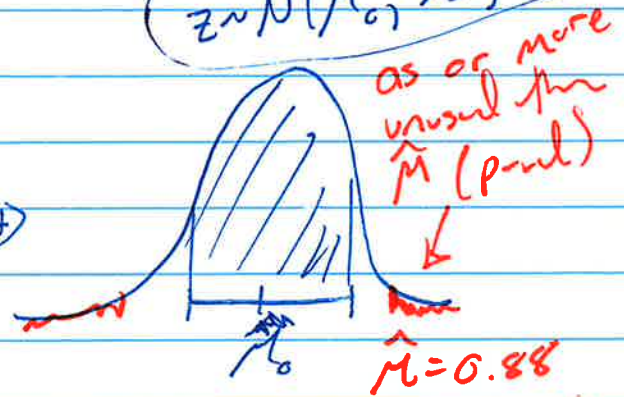
$H_0: \mu = \mu_0 = 0.8$ $\swarrow$ So $\mu = \mu_0$

$H_a: \mu \neq \mu_0 = 0.8$
($>$, $<$)

95% CI



$z \sim N(\mu_0, SE)$



$p\text{-val} < 0.1$ (0.05)

Evidence to reject H_0 . (0.01)

Z-score: $z = \frac{\hat{\mu} - \mu_0}{SE(\mu)}$

$\hookrightarrow N(0, 1)$

(t-test) Small sample: $t_{df} = \frac{\hat{\mu} - \mu_0}{\hat{SE}(\mu)}$, $\hat{SE}(\mu) = \frac{s}{\sqrt{n}}$
 $n = 10$
 $n \leq 30$ $\hookrightarrow t_{df}(0, 1)$

$$Y = f(X) + \varepsilon$$

ps 4

Start inference for simple linear regression

$$\underset{\text{(pop)}}{f(x) = \beta_0 + \beta_1 X} \approx \underset{\text{(sample)}}{\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X}$$

How sure are we of $\beta_0 \approx \hat{\beta}_0$ & $\beta_1 \approx \hat{\beta}_1$?

Need standard errors (SE).

$$SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2} \stackrel{?}{=} \frac{\sigma^2}{\text{Var}(X)}$$

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2} \right), \sigma = \text{Var}(\varepsilon)$$

$$\begin{array}{l} 95\% \text{ CI: } \hat{\beta}_1 \pm 2 SE(\hat{\beta}_1) \\ \hat{\beta}_0 \pm 2 SE(\hat{\beta}_0) \end{array} \left(\begin{array}{l} 95\% \text{ "conf"} \\ \text{true } \beta_0, \beta_1 \\ \text{are in intervals} \end{array} \right)$$

HT in text.